Jasmine Debugger

User Guide

Version 1.0 · Internal Use Only

Desktop diagnostic tool for direct serial access to the Jasmine PCBA (stepper motor + relay controller) via AAEON MTU4 SBC.

1. Overview

Jasmine Debugger establishes an SSH connection to the host SBC, shuts down the active MotorControl process via the ServiceManager HTTP API, then opens a serial relay channel directly to the Jasmine board. Commands are sent and responses received in real time through the structured command panel.

MotorControl must be restarted manually on the SBC after disconnecting. The tool does not restart it automatically.

2. Connection

Enter the connection parameters in the toolbar, then click **Connect**.

Field	Default	Description
`IP`	192.168.1.100	IP address of the AAEON MTU4 SBC
`User`	silentsentinel	SSH username
`Port`	22	SSH port
`Pass`	(password)	SSH password — used for authentication and sudo serial port access
`Baud`	800000	Serial baud rate — must match firmware (800000 for Jasmine)

Connection sequence on **Connect**:

1. SSH connection established to the SBC.
2. MotorControl shut down via ServiceManager HTTP API.
3. Jasmine serial port located under /dev/serial/by-id/ and resolved.
4. Python serial relay launched on the SBC under sudo.
5. Tool becomes ready — command panel is enabled.

Click **Disconnect** to close the relay and SSH session. Restart MotorControl manually on the SBC afterwards.

3. Communication Log

The log on the right records all traffic. Columns: Timestamp, Dir, Data.

Dir	Colour	Meaning
Tx	Blue	Command sent to device
Rx	Green	Response received from device
---	Grey	Connection status message
ERR	Red	Error

Controls: Auto-scroll keeps the latest entry visible. Clear removes all rows. Export Log saves as CSV or TXT. The log holds up to 20,000 rows.

4. Protocol

All commands follow this ASCII format:

`={CMD} [p1 p2 ...]\r\n — set / action`

`=?{CMD} [p1 ...]\r\n — query`

Where 'x' appears in a mnemonic, substitute the axis letter: **P** (pan) or **T** (tilt). For example, MxA → `=MPA 45.0` or `=MTA 45.0`. Joint and system commands do not carry an axis letter.

The **Checksum** checkbox appends an 8-bit arithmetic checksum as two hex digits, e.g. `=MPA 45.0\4F`. Enable if the firmware requires it.

5. Command Panel

The **Axis** selector (P / T) and **Checksum** checkbox at the top apply to all structured commands. A **Manual Send** field at the bottom allows raw command entry.

5.1 Motion Tab

Axis Motion

Command	Parameters	Example	Description
`MxA`	`angle (deg)`	=MPA 45.0	Move axis to absolute position
`MxR`	`offset (deg)`	=MPA 10.0	Move axis by relative offset from current position
`MxJ`	`velocity (deg/s)`	=MPJ 5.0	Jog axis continuously at given velocity; 0 to stop
`MxS`	—	=MPS	Stop axis immediately
`MxC`	`angle (deg)`	=MPC 0.0	Set current position register to given value

Joint Motion

Command	Parameters	Example	Description

MJJ	pan_vel, tilt_vel (deg/s)	=MJJ 5.0 -3.0	Jog both axes simultaneously at independent velocities
MJS	–	=MJS	Stop both axes

Motion Queries

Query	Example	Returns
?MxC	=?MPC	Current axis angle (deg)
?MxE	=?MPE	Raw encoder count
?MxA	=?MPA	Current target position (deg)
?MxS	=?MPS	1 if axis is stationary, 0 if moving
?MxJ	=?MPJ	1 if axis is currently jogging
?MxT	=?MPT	Motor temperature (deg C)
?MxI	=?MPI	Motor current (A)
?MJS	=?MJS	1 if both axes are stationary

5.2 Vel/Profile Tab

Max Velocity

Command	Parameters	Example	Description
MxV	vel (deg/s, 0–9999)	=MPV 60.0	Set maximum velocity for axis motion commands
?MxV	–	=?MPV	Query current max velocity setting

Accel/Decel Profile (MxP) — 8-parameter trapezoidal profile

#	Parameter	Example	Description
f1	Start velocity (deg/s)	0.0	Velocity at which motion begins
f2	Initial acceleration (deg/s²)	100.0	Acceleration rate during initial phase
f3	Initial target vel (deg/s)	30.0	Velocity target at end of initial acceleration
f4	Max acceleration (deg/s²)	200.0	Peak acceleration rate
f5	Max velocity (deg/s)	60.0	Peak cruise velocity
f6	Max deceleration (deg/s²)	200.0	Peak deceleration rate
f7	Stop deceleration (deg/s²)	100.0	Final deceleration rate approaching stop

`f8`	`Release velocity (deg/s)`	0.0	Velocity below which motion is considered complete

Example: `=MPP 0.0 100.0 30.0 200.0 60.0 200.0 100.0 0.0` — use **Get ?MxP** to read back the active profile.

Position Streaming (MxU)

Command	Parameters	Example	Description
`MxU`	`period (ms, 0–10000)`	=MPU 100	Set interval for unsolicited position updates; 0 disables streaming
`?MxU`	—	=?MPU	Query current streaming period

5.3 Config Tab

Axis Limits (MxL)

Command	Parameters	Example	Description
`MxL`	`lower, upper (deg)`	=MPL -170.0 170.0	Set software travel limits; motion commands are clamped to this range
`?MxL`	—	=?MPL	Query current limits

Correction Sensitivity (MxD)

Command	Parameters	Example	Description
`MxD`	`mismatch, stall (deg)`	=MPD 1.0 5.0	Mismatch: max encoder/commanded error before correction; Stall: stall detection threshold
`?MxD`	—	=?MPD	Query current sensitivity settings

Encoder

Command	Parameters	Example	Description
`MxX`	`ticks (int, 1–999999)`	=MPX 4096	Encoder resolution in counts per revolution
`?MxX`	—	=?MPX	Query encoder resolution
`MxW`	`constant (float)`	=MPW 0.0879	Encoder scaling constant (degrees per tick)
`?MxW`	—	=?MPW	Query scaling constant

Axis Flags (MxF)

Command	Parameters	Example	Description

`MxF`	`flags (int, 0–255)`	=MPF 3	Bitmask of axis behaviour flags; refer to firmware documentation for bit definitions

5.4 Homing Tab

Run Homing

Command	Parameters	Example	Description
`MJH`	`[debug 0/1]`	=MJH or =MJH 1	Home both axes; optional debug flag enables verbose output
`?MJH`	–	=?MJH	Query current homing status
`MxH`	`mode (0–6)`	=MPH 3	Home selected axis using specified mode (see table below)
`?MxH`	–	=?MPH	Query current homing mode

Value	Mode	Description
`0`	Default	Firmware default homing behaviour
1	Manual	Software-triggered home position
2	Line	Line sensor-based homing
3	Limits	Travel-to-limit homing
4	Strip	Optical strip sensor homing
5	Notch	Notch/slot sensor homing
6	Zline	Z-index line homing

Homing Velocities (MJV)

Command	Parameters	Example	Description
`MJV`	`home_vel, post_vel [, self_corr_vel] (deg/s)`	=MJV 10.0 5.0	Home search velocity, post-trigger settling velocity, optional self-correction velocity
`?MJV`	–	=?MJV	Query current homing velocity settings

Homing Config

Command	Parameters	Example	Description
`MxO`	`offset (deg)`	=MPO 0.0	Angular offset applied to position register after homing completes
`?MxO`	–	=?MPO	Query post-home offset
`MxK`	`temp_offset (deg C)`	=MPK 0.0	Temperature compensation offset applied during homing
`?MxK`	–	=?MPK	Query temperature compensation offset
`HDE`	`delay (s, 0–9999)`	=HDE 0	Delay before homing sequence begins

?HDE	–	=?HDE	Query homing delay

5.5 System Tab

System Actions

Command	Description
SL!	Save current settings to non-volatile memory
SLD	Restore factory defaults (does not save automatically — follow with SL! if needed)
SLR	System reset; ?SLR returns uptime
SLB	Reset communication buffers; ?SLB returns buffer status
?SLV	Query firmware version string
MJR	Reset joint motion controller
MJD	Disable joint motion

System Settings

Command	Parameters	Example	Description
SLM	model (12/41/42)	=SLM 42	Set product model: 12=AERON, 41=JAEGAR V1, 42=JAEGAR V2
?SLM	–	=?SLM	Query model number
SLS	serial_no (0–99999)	=SLS 1234	Set unit serial number
?SLS	–	=?SLS	Query serial number
SLF	flags (int, 0–65535)	=SLF 0	System-level control flags bitmask
?SLF	–	=?SLF	Query control flags
SLT	temp (deg C)	=SLT 25.0	Override reported system temperature
?SLT	–	=?SLT	Query system temperature
?SLH	–	=?SLH	Query ambient humidity (%)
?SLI	–	=?SLI	Query IMU data
?SEV	–	=?SEV	Query recent event log

Cold Start

Command	Parameters	Example	Description
SLZ	bits (int, 0–1023)	=SLZ 0	Heater configuration bitmask

?SLZ	–	=?SLZ	Query heater config
SLC	threshold (deg C)	=SLC 5.0	Temperature below which cold-start behaviour activates
?SLC	–	=?SLC	Query cold threshold
SLW	temp (deg C)	=SLW 20.0	Keep-warm target temperature
?SLW	–	=?SLW	Query keep-warm temperature

Network (SLN)

Command	Parameters	Example	Description
SLN	ip, gateway, mask, port	=SLN 192.168.1.100 192.168.1.1 255.255.255.0 8080	Set network interface configuration
?SLN	–	=?SLN	Query current network settings

5.6 Power/I-O Tab

Power Channels (PxC, x = 0–3)

Command	Parameters	Example	Description
PxC	state (0/1)	=P1C 1	Enable (1) or disable (0) power channel x
?PxC	–	=?P1C	Query power channel state
P2V	voltage (V, 0–60)	=P2V 12.0	Set output voltage on channel 2 (programmable channel only)
?PxV	–	=?P1V	Query voltage on channel x (V)
?PxI	–	=?P1I	Query current draw on channel x (A)
?PxT	–	=?P1T	Query temperature on channel x (deg C)

Motor Current (TMC5160 IRUN / IHOLD)

Command	Parameters	Example	Description
SIR	irun (int, 0–31)	=SIR 20	Set motor run current (TMC5160 IRUN); higher = more current
SIH	ihold (int, 0–31)	=SIH 8	Set motor hold current (TMC5160 IHOLD); typically lower than IRUN

Half Bridge (BcX, c = 1–4)

Command	Parameters	Example	Description
BcC	state (0/1)	=B1C 1	Enable (1) or disable (0) half-bridge channel c
?BcC	–	=?B1C	Query half-bridge channel state

`BcD`	`duty (int, 0–100)`	=B1D 50	Set PWM duty cycle on half-bridge channel c (%)
`?BcD`	–	=?B1D	Query duty cycle
`?BcI`	–	=?B1I	Query current through half-bridge channel c (A)
`?BcV`	–	=?B1V	Query voltage on half-bridge channel c (V)
`?BcT`	–	=?B1T	Query temperature of half-bridge channel c (deg C)

Optically Isolated (OcX, c = 1–2)

Command	Parameters	Example	Description
`OcC`	`state (0/1)`	=O1C 1	Enable (1) or disable (0) opto-isolated output channel c
`?OcC`	–	=?O1C	Query channel state
`OcD`	`duty (int, 0–100)`	=O1D 75	Set PWM duty cycle on opto-isolated channel c (%)
`?OcD`	–	=?O1D	Query duty cycle

Fan Control (F1x)

Command	Parameters	Example	Description
`F1C`	`state (0/1)`	=F1C 1	Enable (1) or disable (0) fan
`?F1C`	–	=?F1C	Query fan enable state
`F1D`	`duty (int, 0–100)`	=F1D 50	Set fan PWM duty cycle (%)
`?F1D`	–	=?F1D	Query current fan duty cycle
`?F1V`	–	=?F1V	Query fan supply voltage (V)
`?F1I`	–	=?F1I	Query fan current (A)

XIO GPIO (XcS, c = 0–7)

Command	Parameters	Example	Description
`XcS`	`state (0/1)`	=X3S 1	Set GPIO pin c high (1) or low (0)
`?XcS`	–	=?X3S	Query current state of GPIO pin c

5.7 Debug Tab

TMC5160 Register (MxM)

Direct read/write access to the TMC5160 stepper driver register file. Incorrect register values can affect motor behaviour or cause fault conditions. Refer to the TMC5160 datasheet for register addresses and field definitions.

Command	Parameters	Example	Description
`MxM`	`addr (0–255), value (int)`	=MPM 0 0	Write value to TMC5160 register at address addr on selected axis
`?MxM`	`addr (0–255)`	=?MPM 0	Read current value of TMC5160 register at address addr

Debug Actions

Command	Description
`PSE`	Print all current settings to the communications log
`RSE`	Re-read and refresh settings from EEPROM
`PMD`	Print motor debug dump (driver state, currents, flags)
`SFT <temp>`	Set a fake temperature (deg C) for testing; values outside –100 to 200 disable the override

5.8 Manual Send

The text field at the bottom of the command panel sends raw text directly to the serial port. `\r\n` is appended automatically if not already present. Use for commands not covered by the structured panel, one-off experiments, or entering multi-parameter commands by hand.

6. Notes

- Settings changed via commands are **not persisted** unless `SL!` (Save) is issued.
- After disconnecting, **MotorControl must be restarted manually** on the SBC.
- Correct baud rate for this hardware is **800000**.